

Consumer Benchmark Comparison Dashboard

Interactive survey reporting tool for category, segment, and total-market benchmark comparison

Survey analytics	Benchmarking	Tableau	Python	Weighted reporting	Client storytelling
------------------	--------------	---------	--------	--------------------	---------------------

Executive summary

Designed a Tableau survey reporting dashboard to help research and client-facing teams compare category performance against segment and total-market benchmarks. The project replaced slower static crosstab/manual comparisons with a more flexible tool for exploring purchase motives, attitudes, rankings, demographics, and benchmark summaries.

This reduced repetitive benchmark-checking work and gave research/client teams a faster way to identify category-level stories before building client deliverables.

Business problem

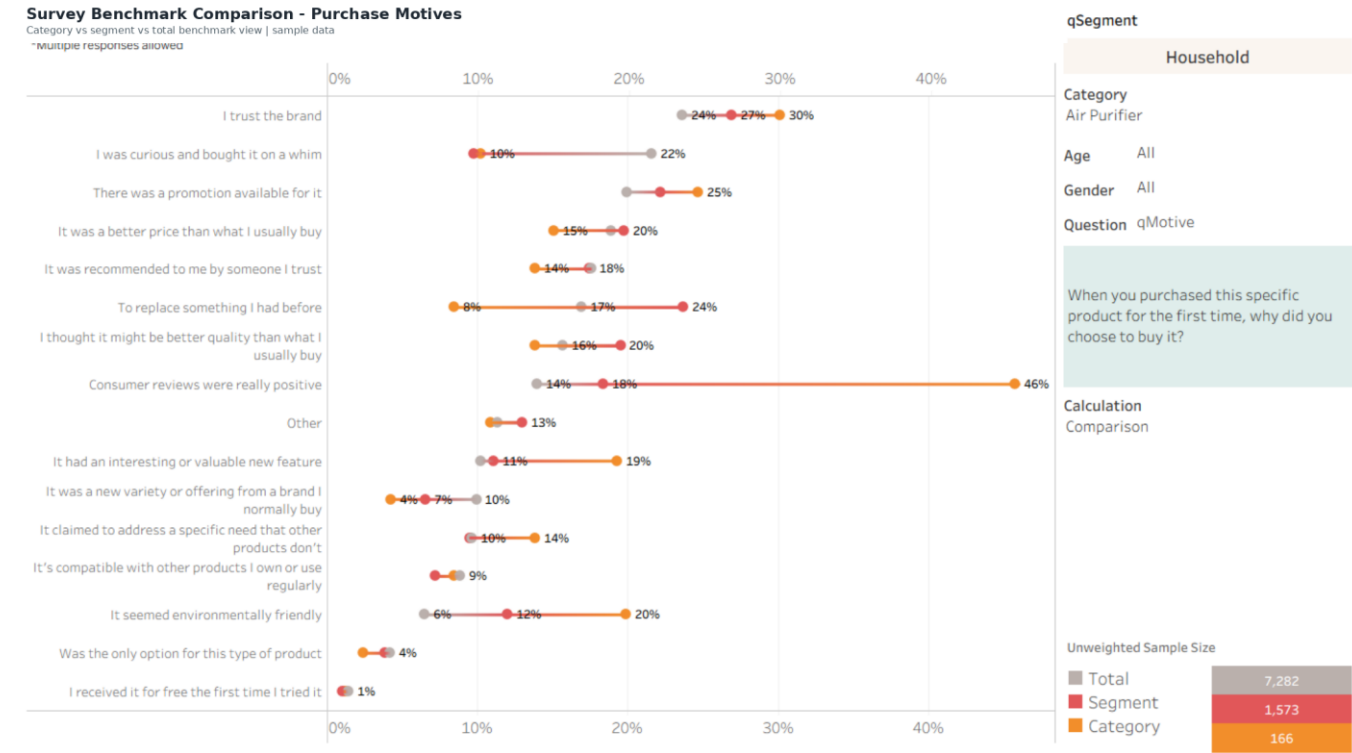
Survey data captured purchase drivers, attitudes, satisfaction, and demographics, but raw outputs were slow to compare across category, segment, and total market.

What I built

A dashboard-ready survey data structure and Tableau views for weighted comparisons, metric rollups, rankings, demographics, and benchmark summaries.

Portfolio value

Shows survey analytics beyond charts: data reshaping, flexible metric logic, context-aware benchmark comparison, and client-friendly interpretation.



Role & contribution

- Restructured looped survey data into a stacked, dashboard-ready format using Python
- Designed Tableau views for Total vs Segment vs Category comparison
- Built flexible calculation logic for metric types, filters, and benchmark views

Confidentiality note: visuals are presented as sanitized/recreated portfolio examples. Project-specific acronyms and client/employer identifiers should be removed or generalized for public sharing.

Tools & methods

- Tableau calculated fields and LOD expressions for stable filtering and comparisons
- Python for survey data preparation and reshaping
- Weighted reporting logic, T2B/B2B rollups, average-based metrics, and sample-size visibility

1. Business problem and reporting logic

Business problem

The survey captured product purchase behavior, brand choice, purchase drivers, satisfaction, and related attitudes. However, the raw outputs were not designed for quick comparison or client-ready storytelling.

- How is a category performing relative to its segment?
- Does the story change when compared with the total market?
- Which drivers or attitudes are strongest for a given category?
- How do results differ in direct comparison versus difference-from-total views?

Design challenge

The dashboard needed to support flexible exploration without making the logic hard to trust. The key design problem was balancing flexibility, stable calculation behavior, and interpretability.

- Support category, segment, and total comparisons in one framework
- Handle multiple question types and reporting styles
- Keep results consistent as filters changed
- Make sample-size context visible for interpretation

Core analytical idea

A category can look strong against the total market but average within its own segment. Making that distinction visible helps stakeholders avoid overclaiming and frame performance in the right competitive context.

Data preparation and metric logic

Wide-to-stacked reshape

Respondent-level records contained product-loop entries, so Python was used to normalize them into a dashboard-ready structure.

Multi-level comparison

The reporting layer supported Total, Segment, and Category views within the same dashboard framework.

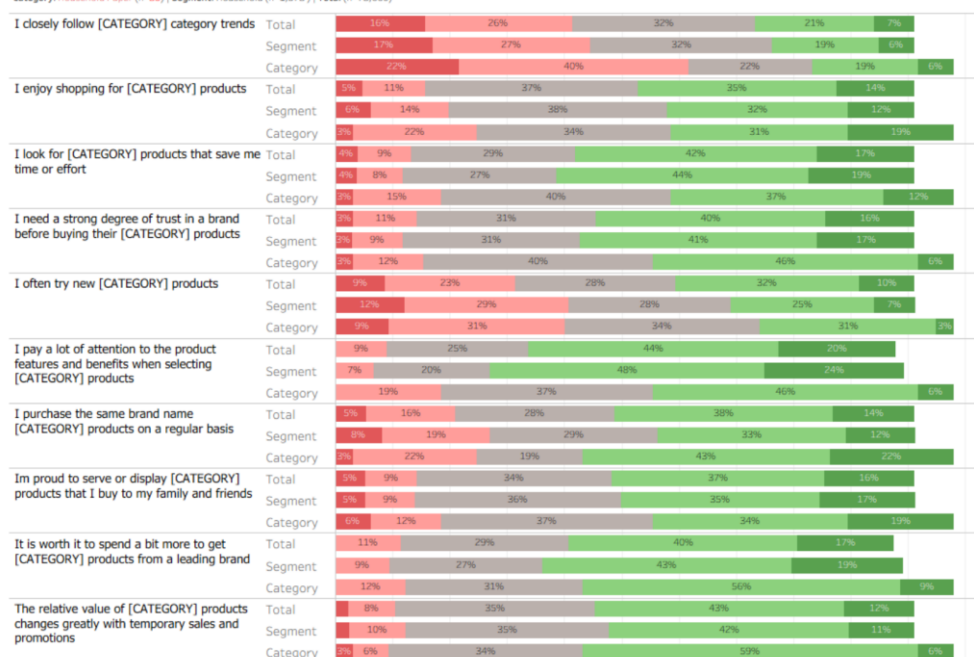
Metric flexibility

The dashboard handled average-based metrics, Top-2-Box / Bottom-2-Box rollups, direct comparison, and difference-from-total logic.

Survey Benchmark Comparison - Category Attitudes

Likert-scale view with category, segment, and total comparisons | sample data

Survey Segment: qSegment | Question: qCategoryAttitudes | Likert Scale: 5 | Weight: 1 | Filter: All



Likert-style view: attitude statements compared across total, segment, and category with scale distribution visible in one screen.

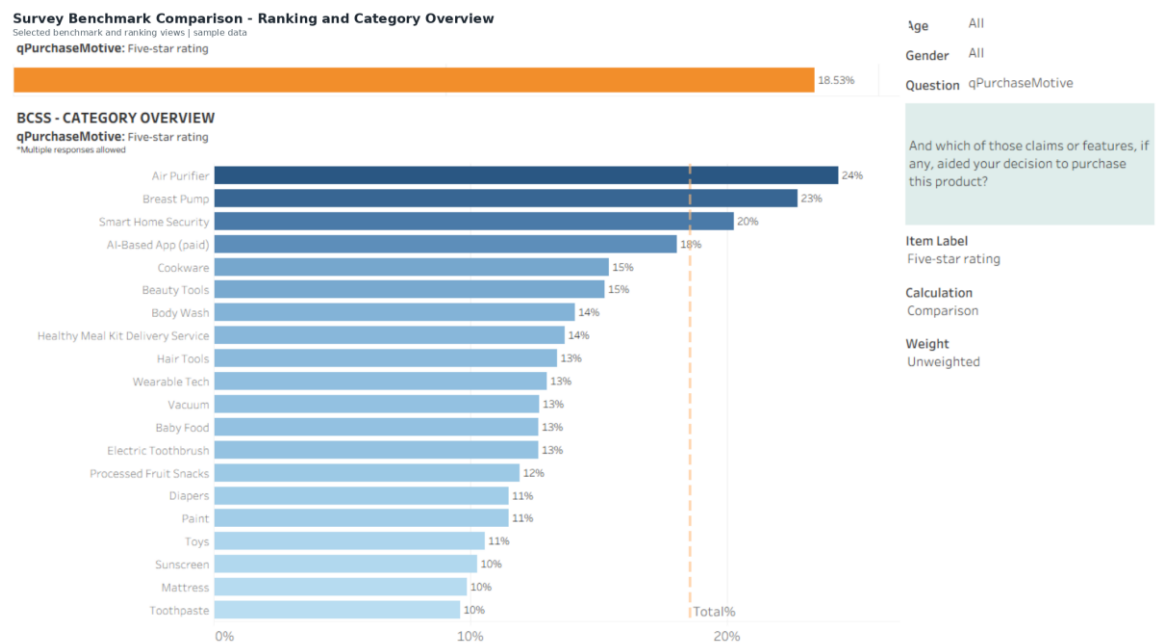
2. Dashboard structure and benchmark views

Dashboard views

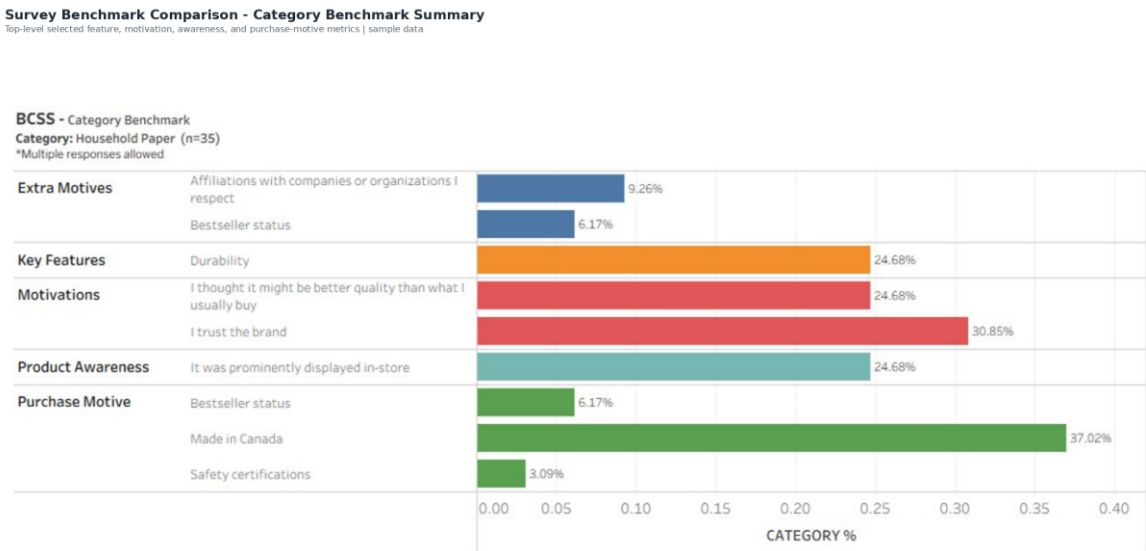
- Question view for multi-select survey questions such as motives, awareness, and exploratory measures
- Likert view for scale-based reporting
- Ranking view to show category standing based on selected question values
- Demographic view to explore audience profile cuts
- Benchmark view to summarize top-level performance across key questions

Key design features

- Total vs Segment vs Category comparisons in a single view
- Weighted and unweighted reporting logic for interpretation and QA
- Interactive filters for category, demographic group, and question selection
- Sample-size visibility to make small category cuts easier to interpret
- Exploratory navigation from broad benchmark summary to focused category detail



Ranking and category overview views: selected benchmark metrics and category standing shown alongside total-average reference points.

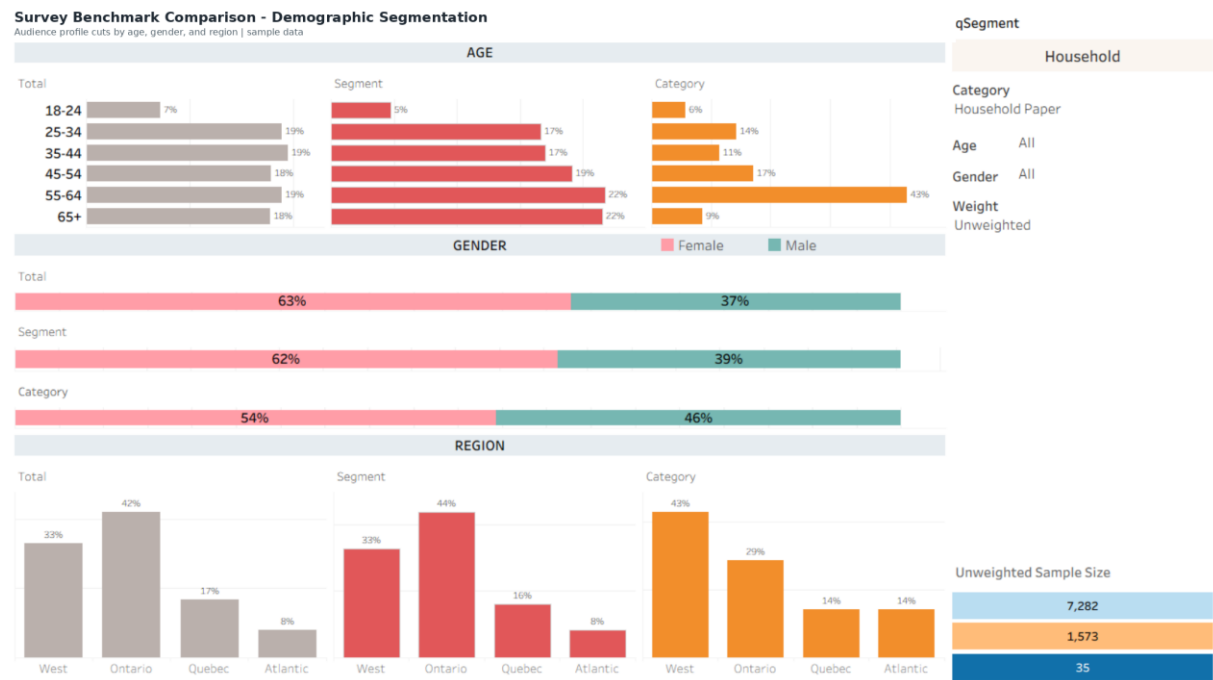


Benchmark summary view: selected metrics grouped across key features, motivations, awareness, and purchase motives.

Design judgment

The dashboard was not intended to replace analyst interpretation. It was designed to reduce repetitive manual comparison, make benchmark context easier to explore, and help stakeholders identify which differences deserved deeper explanation.

3. Project value, limitations, and portfolio takeaways



Demographic segmentation view: audience profile cuts by age, gender, and region, with total, segment, and category context.

Business value

- Reduced reliance on static crosstabs and manual comparisons
- Made category, segment, and total-market benchmark comparisons faster to explore
- Supported client-facing storytelling by showing whether category performance was truly differentiated within context
- Enabled self-serve exploration across questions, demographics, and benchmark summaries

Limitations / next improvements

- Add statistical significance or confidence indicators for category vs segment differences
- Add stronger sample-size warnings for small category cuts
- Create export-ready views for client presentations
- Add guided insight annotations so non-technical users know what to look for

Project takeaway

This project demonstrates how survey reporting can move beyond static tables. By combining data reshaping, flexible calculation logic, benchmark comparison, and exploratory dashboard design, the output made category performance easier to interpret and easier to package for stakeholder discussion.

What this project demonstrates

- Survey data preparation and reshaping
- Weighted reporting and multiple metric types
- Benchmark comparison across total, segment, and category
- Client-friendly interpretation and self-serve exploration